ASSIGNMENT 15.5

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Batch 20

Task 1 – Student Attendance API Task:
Create a RESTful API for student attendance tracking.
Instructions:
• Endpoints required: o GET /attendance → View attendance records o POST /attendance → Mark attendance o PUT /attendance/{id} → Update attendance record o DELETE /attendance/{id} → Remove attendance entry • Ensure JSON-based request and response handling. Expected Output:
• API for maintaining and updating attendance records.

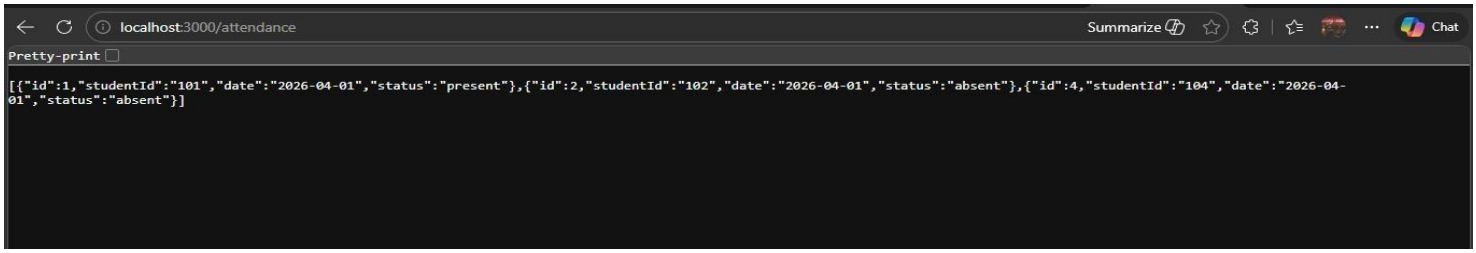
CODE:

```
const express = require('express'); const
app = express();

// Middleware to parse incoming JSON requests  app.use(express.json());

// In-memory storage for attendance records let attendanceRecords
= [];
let nextId = 1;

// Root route to confirm the API is running  app.get('/',
(req, res) => {    res.status(200).send('Attendance API is
running');
});

// GET /attendance - return all attendance records app.get('/attendance', (req, res)
=> {   res.status(200).json(attendanceRecords);
});

// POST /attendance - add a new attendance record
app.post('/attendance', (req, res) => {   const { studentId,
date, status } = req.body;

  if (!studentId || !date || !status) {      return res.status(400).json({       message:
'Missing required fields: studentId, date, and status are all required',
    });
  }
```

```
  const normalizedStatus = status.toString().trim().toLowerCase();   if
(normalizedStatus !== 'present' && normalizedStatus !== 'absent') {     return
res.status(400).json({
     message: "Status must be either 'present' or 'absent'",
    });
  }

  const newRecord = {
id: nextId++,     studentId,
    date,      status:
normalizedStatus,
  };

  attendanceRecords.push(newRecord);
res.status(201).json(newRecord);
});

// PUT /attendance/:id - update an existing attendance record  app.put('/attendance/:id',
(req, res) => {   const
recordId = Number(req.params.id);   const { studentId,
date, status } = req.body;

  const record = attendanceRecords.find((item) => item.id === recordId);   if
(!record) {      return res.status(404).json({ message: 'Attendance record
not found' });
  }

  if (!studentId || !date || !status) {
return res.status(400).json({
     message: 'Missing required fields: studentId, date, and status are all required',
    });
  }

  const normalizedStatus = status.toString().trim().toLowerCase();   if
(normalizedStatus !== 'present' && normalizedStatus !== 'absent') {     return
res.status(400).json({
     message: "Status must be either 'present' or 'absent'",
    });
  }
 record.studentId = studentId;
record.date = date;
  record.status = normalizedStatus;

  res.status(200).json(record);
});
```

```
// DELETE /attendance/:id - remove an attendance record  app.delete('/attendance/:id',
(req, res) => {   const
recordId = Number(req.params.id);
  const index = attendanceRecords.findIndex((item) => item.id === recordId);

  if (index === -1) {     return res.status(404).json({ message: 'Attendance
record not found' });
  }

  const deletedRecord = attendanceRecords.splice(index, 1)[0];   res.status(200).json(deletedRecord);
});

// Start Express server  const PORT = process.env.PORT || 3000;
app.listen(PORT, () => {    console.log(`Attendance API server is
running on port ${PORT}`);
});
```

OUTPUT:



Task 2 – Inventory Management API Task:
Develop a REST API for warehouse inventory management.
Instructions:
• Endpoints required: o GET /inventory → List all items o POST /inventory → Add new item o PUT /inventory/{id} → Update stock quantity o DELETE /inventory/{id} → Remove an item • Implement validation for stock values.
Expected Output:
• API capable of managing inventory stock levels.

PROMPT:
//Create a RESTful API for warehouse inventory management using Node.js and Express. The API should handle JSON requests and responses and store data in an in-memory array. Each inventory item must include id, itemName, quantity, and price. Implement the following endpoints: GET /inventory to return all items, POST /inventory to add a new item with validation (itemName required, quantity must be a non-negative number, price must be positive), PUT /inventory/:id to update the stock quantity of an existing item with proper validation and error handling if the item is not found, and DELETE /inventory/:id to remove an item. Include a root route GET / that returns a simple message like

'Inventory API is running'. Use express.json() middleware, proper HTTP status codes (200, 201, 400, 404), and add clear comments so the code is easy to understand and ready to run in a single server.js file

CODE:

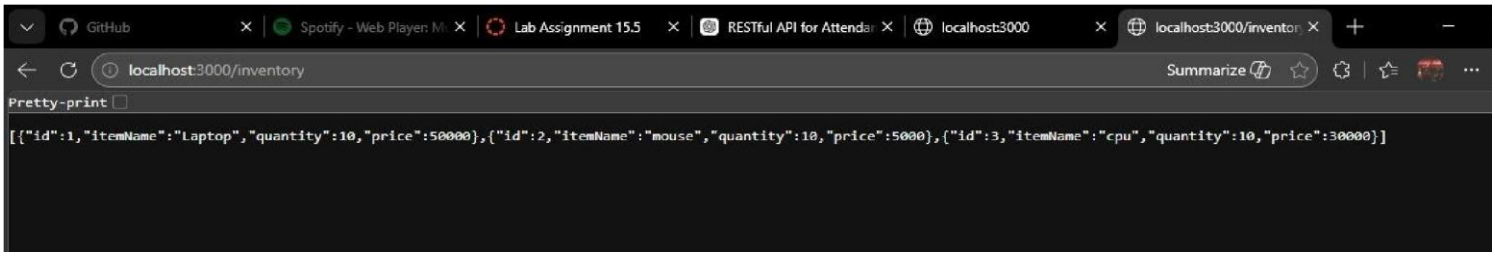
```
console.log("Inventory Server starting...");

const express = require('express'); const
app = express();

app.use(express.json());

let inventory = [];

// Home route  app.get('/', (req, res) => {
res.send("Inventory API is running...");
});

// GET /inventory → List all items app.get('/inventory',
(req, res) => {     res.json(inventory);
});

// POST /inventory → Add new item app.post('/inventory',
(req, res) => {      const { itemName, quantity,
price } = req.body;

   // Validation      if (!itemName || quantity === undefined || price
=== undefined) {
return res.status(400).json({ message: "All fields are required" });
   }

  if (quantity < 0 || price <= 0) {       return res.status(400).json({ message: "Invalid
stock or price value" });
   }

   const newItem = {        id:
inventory.length + 1,
itemName,        quantity,        price
   };

   inventory.push(newItem);
res.status(201).json(newItem);  });

// PUT /inventory/:id → Update stock quantity
app.put('/inventory/:id', (req, res) => {     const itemId
= parseInt(req.params.id);
```

```
   const { quantity } = req.body;

   if (quantity === undefined || quantity < 0) {         return
res.status(400).json({ message: "Invalid quantity value" });
   }

   const item = inventory.find(i => i.id === itemId);

   if (!item) {         return res.status(404).json({ message: "Item
not found" });
   }

   item.quantity = quantity;

   res.json(item);
});

// DELETE /inventory/:id → Remove item app.delete('/inventory/:id',
(req, res) => {      const itemId =
parseInt(req.params.id);

   const index = inventory.findIndex(i => i.id === itemId);

   if (index === -1) {         return res.status(404).json({ message:
"Item not found" });
   }

  const deletedItem = inventory.splice(index, 1);

   res.json(deletedItem[0]);
});

// Start server const
PORT = 3000;

app.listen(PORT, () => {      console.log("Server is
running on port " + PORT);
});
```

OUTPUT:

Task 3 – Hotel Room Booking API Task:
Develop a RESTful API for hotel room booking.
Instructions:

- Endpoints required: o GET /rooms → View available rooms o POST /rooms/book → Book a room o GET /rooms/{id} →



View booking details o DELETE /rooms/{id} → Cancel booking

- Implement error handling for unavailable rooms. Expected Output:
- API simulating hotel room booking operations.

PROMPT:

//Create a RESTful API for a hotel room booking system using Node.js and Express. The API should handle JSON requests and responses and store data in an in-memory array. Each room should have fields like id, roomNumber, type, price, and isBooked (boolean). Implement the following endpoints: GET /rooms to list all available rooms (only those not booked), POST /rooms/book to book a room by providing room id and customer details, GET /rooms/:id to view booking details of a specific room, and DELETE /rooms/:id to cancel a booking and mark the room as available again. Include proper error handling such as returning a message if a room is already booked or not found. Use express.json() middleware, appropriate HTTP status codes (200, 201, 400, 404), and include a root route GET / that returns 'Hotel Booking API is running'.

Ensure the code is clean, well-commented, and ready to run in a single server.js file.

CODE:

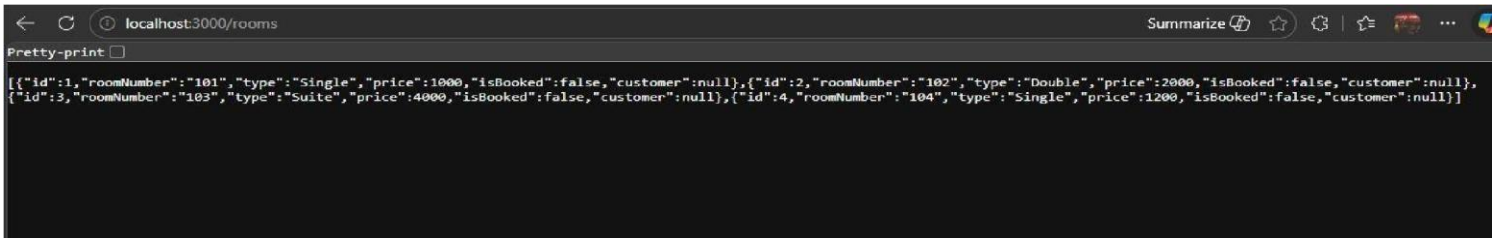
```
console.log("Hotel Booking Server starting...");

const express = require('express'); const
app = express();

app.use(express.json());

// Sample rooms data let rooms
= [
  { id: 1, roomNumber: "101", type: "Single", price: 1000, isBooked: false, customer: null },
  { id: 2, roomNumber: "102", type: "Double", price: 2000, isBooked: false, customer: null },
  { id: 3, roomNumber: "103", type: "Suite", price: 4000, isBooked: false, customer: null },
  { id: 4, roomNumber: "104", type: "Single", price: 1200, isBooked: false, customer: null }
];

// Home route  app.get('/',
(req, res) => {
res.send("Hotel Booking
API is running...");
});

// GET /rooms → View available rooms app.get('/rooms',
```

```
(req, res) => {     const availableRooms = rooms.filter(room => !room.isBooked);
res.json(availableRooms);
});

// POST /rooms/book → Book a room
app.post('/rooms/book', (req, res) => {     const { roomId,
customerName } = req.body;

   if (!roomId || !customerName) {         return res.status(400).json({ message: "roomId
and customerName are required" });
   }

   const room = rooms.find(r => r.id === roomId);

   if (!room) {         return res.status(404).json({ message: "Room
not found" });
   }

   if (room.isBooked) {         return res.status(400).json({ message:
"Room already booked" });
   }

   room.isBooked = true;     room.customer
= customerName;

   res.status(201).json({         message:
"Room booked successfully",
room
   });
});

// GET /rooms/:id → View booking details
app.get('/rooms/:id', (req, res) => {     const roomId
= parseInt(req.params.id);
   const room = rooms.find(r => r.id === roomId);

   if (!room) {         return res.status(404).json({ message: "Room
not found" });
   }

   res.json(room);
});

// DELETE /rooms/:id → Cancel booking
app.delete('/rooms/:id', (req, res) => {     const roomId
```

```
= parseInt(req.params.id);     const room =
rooms.find(r => r.id === roomId);

   if (!room) {         return res.status(404).json({ message: "Room
not found" });
   }

   if (!room.isBooked) {         return res.status(400).json({ message:
"Room is not booked" });
   }

   room.isBooked = false;
room.customer = null;

   res.json({         message: "Booking cancelled
successfully",         room
   });
});

// Start server const
PORT = 3000;

app.listen(PORT, () => {      console.log("Server is
running on port " + PORT);
});
```

OUTPUT:

Task 4 – Online Voting API Task:
Create a RESTful API for an online voting system.



Instructions:

- Endpoints required: o GET /candidates → List candidates o POST /vote → Cast a vote o GET /results → View voting results
- Ensure one vote per user using in-memory validation.
  Expected Output:
- API simulating a secure online voting process.

PROMPT:

//Design and implement a RESTful API for an online voting system that allows users to participate in a secure and controlled voting process. The API should include endpoints to retrieve a list of candidates (GET /candidates), cast a vote (POST /vote), and view the current voting results (GET /results). The system must enforce a strict rule of one vote per user by using in-memory validation to track users who have already voted and prevent duplicate submissions. Proper error handling should be implemented for cases such as invalid candidate IDs or repeated voting attempts. The final API should simulate a simple yet secure online voting environment, ensuring fairness and correctness in vote counting.

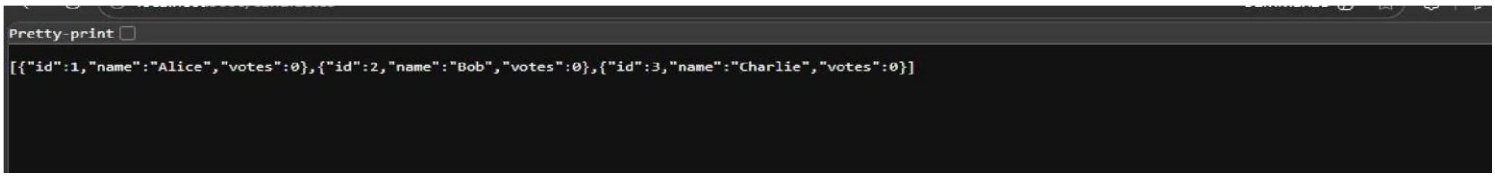
CODE:

```
const express = require('express'); const
app = express(); app.use(express.json());

// In-memory data let candidates
= [
   { id: 1, name: "Alice", votes: 0 },
   { id: 2, name: "Bob", votes: 0 },
   { id: 3, name: "Charlie", votes: 0 }
];

// Store users who have voted  let
votedUsers = new Set();

// GET /candidates → List candidates app.get('/candidates', (req,
res) => {     res.json(candidates);
});

// POST /vote → Cast a vote app.post('/vote',
(req, res) => {      const { userId,
candidateId } = req.body;

   // Validate input     if
(!userId || !candidateId) {         return res.status(400).json({ message: "userId and
candidateId are required" });
   }

   // Check if user already voted     if
(votedUsers.has(userId)) {       return res.status(403).json({ message:
 "User has already voted" });
   }

   // Find candidate      const candidate = candidates.find(c => c.id
=== candidateId);     if
```

```
(!candidate) {        return res.status(404).json({ message:
"Candidate not found" });
   }

   // Cast vote     candidate.votes
+= 1;
votedUsers.add(userId);

   res.json({ message: "Vote cast successfully" });
});

// GET /results → View results  app.get('/results', (req,
res) => {     res.json(candidates);
});

// Start server const PORT
= 3000; app.listen(PORT,
() => {
   console.log(`Server running on port ${PORT}`);
});
```

OUTPUT:

Task 5 – Teacher Management API Task:



Develop a RESTful API to manage teacher records.
Instructions:

- Endpoints required: o GET /teachers → List all teachers o POST /teachers → Add a new teacher o PUT /teachers/{id} → Update teacher details o DELETE /teachers/{id} → Remove a teacher • Ensure proper error handling and JSON responses. Expected Output:
- Functional API for managing teacher information.

PROMPT:

//Design and implement a RESTful API for an online voting system that allows users to participate in a secure and controlled voting process. The API should include endpoints to retrieve a list of candidates (GET /candidates), cast a vote (POST /vote), and view the current voting results (GET /results). The system must enforce a strict rule of one vote per user by using in-memory validation to track users who have already voted and prevent duplicate submissions. Proper error handling should be implemented for cases such as invalid candidate IDs or repeated voting attempts. The final API should simulate a simple yet secure online voting environment, ensuring fairness and correctness in vote counting.

CODE:

```
const express = require('express'); const
app = express(); app.use(express.json());

// In-memory data let candidates
= [
   { id: 1, name: "Alice", votes: 0 },
   { id: 2, name: "Bob", votes: 0 },
   { id: 3, name: "Charlie", votes: 0 }
];

// Store users who have voted  let
votedUsers = new Set();

// GET /candidates → List candidates app.get('/candidates', (req,
res) => {     res.json(candidates);
});

// POST /vote → Cast a vote app.post('/vote',
(req, res) => {      const { userId,
candidateId } = req.body;

   // Validate input     if
(!userId || !candidateId) {         return res.status(400).json({ message: "userId and
candidateId are required" });
   }

   // Check if user already voted     if
(votedUsers.has(userId)) {         return res.status(403).json({ message:
"User has already voted" });
   }

   // Find candidate      const candidate = candidates.find(c => c.id
=== candidateId);     if
(!candidate) {         return res.status(404).json({ message:
"Candidate not found" });
   }

   // Cast vote     candidate.votes
+= 1;     votedUsers.add(userId);

   res.json({ message: "Vote cast successfully" });
});
```

```
// GET /results → View results app.get('/results',
(req, res) => {     res.json(candidates);
});

// Start server const PORT = 3000;
app.listen(PORT, () => {      console.log(`Server
running on port ${PORT}`);
});
```

OUTPUT: